Sheth L.U.J & Sir M.V College of Science
Subject :- Artificial Intelligence
Practical no 3

Aim :- Feed Forward Backpropagation Neural Network

1. **Implement the Feed Forward Backpropagation algorithm to train a neural network.**
2. **Use a given dataset to train the neural network for a specific task.**
3. **Evaluate the performance of the trained network on test data.**

Description :-

Step	Code / Function	Description
1	**pd.read_csv()**	Loads the Student Performance dataset.
2	**df.dropna()**	Removes rows containing missing values.
3	**performance_category()**	Converts exam scores into Low, Medium, and High categories.
4	**X and y**	Separates input features and target variable.
5	**pd.get_dummies()**	Converts categorical data into numerical values.
6	**LabelEncoder()**	Converts target classes into numerical labels.
7	**train_test_split()**	Divides data into 80% training and 20% testing data.
8	**StandardScaler()**	Scales the input features for better neural-network training.
9	**MLPClassifier()**	Creates the Feed Forward Neural Network with hidden layers.
10	**model.fit()**	Trains the neural network using backpropagation.
11	**model.predict()**	Predicts student performance on test data.

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12	**accuracy_score()**	Calculates the accuracy of the trained model.
13	**classification_report()**	Displays precision, recall, and F1-score.
14	**confusion_matrix()**	Shows correct and incorrect predictions.
15	**plt.imshow()**	Visualizes the confusion matrix.
16	**model.loss_curve_**	Shows the training loss during neural-network learning.
17	**plt.plot()**	Visualizes the training loss curve.

Code :-

```
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score
from sklearn.metrics import classification_report
from sklearn.metrics import confusion_matrix

df = pd.read_csv("StudentPerformanceFactors.csv")

print("Dataset Loaded Successfully!")

print("\nDataset Shape:")
print(df.shape)

print("\nFirst 5 Rows:")
print(df.head())


df = df.dropna()

print("\nMissing values removed!")


def performance_category(score):

    if score < 60:
        return "Low"
```

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```python
    elif score < 80:
        return "Medium"

    else:
        return "High"


df["Performance_Category"] = df["Exam_Score"].apply(
    performance_category
)

X = df.drop(
    ["Exam_Score", "Performance_Category"],
    axis=1
)

y = df["Performance_Category"]


X = pd.get_dummies(
    X,
    drop_first=True
)

encoder = LabelEncoder()

y = encoder.fit_transform(y)
```

```python
print("\nClasses:")

for i, name in enumerate(encoder.classes_):
    print(i, "=", name)

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)


print("\nTraining Samples:")
print(len(X_train))

print("\nTesting Samples:")
print(len(X_test))

scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)

X_test = scaler.transform(X_test)


model = MLPClassifier(
    hidden_layer_sizes=(32, 16),
```

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T092 Practical no 4.py - C:/Users/Vinod Mali/Desktop/Ai/T092 Practical no 4.py (3.14.6)

```
    activation="relu",
    solver="adam",
    max_iter=500,
    random_state=42
)


model.fit(
    X_train,
    y_train
)

print("\nNeural Network trained successfully!")


y_pred = model.predict(X_test)


accuracy = accuracy_score(
    y_test,
    y_pred
)

print("\nAccuracy:")
print(accuracy)

print("\nAccuracy Percentage:")
print(round(accuracy * 100, 2), "%")
```

T092 Practical no 4.py - C:/Users/Vinod Mali/Desktop/Ai/T092 Practical no 4.py (3.14.6)

```
print("\nClassification Report:")

print(
    classification_report(
        y_test,
        y_pred,
        target_names=encoder.classes_,
        zero_division=0
    )
)


cm = confusion_matrix(
    y_test,
    y_pred
)

print("\nConfusion Matrix:")
print(cm)


plt.figure(figsize=(7, 6))

plt.imshow(cm)

plt.title(
    "Confusion Matrix - Neural Network"
)
```

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```
plt.xlabel(
    "Predicted Class"
)

plt.ylabel(
    "Actual Class"
)

plt.xticks(
    range(len(encoder.classes_)),
    encoder.classes_
)

plt.yticks(
    range(len(encoder.classes_)),
    encoder.classes_
)


for i in range(len(cm)):

    for j in range(len(cm)):

        plt.text(
            j,
            i,
            cm[i, j],
            ha="center",
            va="center"
```

```
        )


plt.colorbar()

plt.show()


plt.figure(figsize=(8, 5))

plt.plot(
    model.loss_curve_
)

plt.title(
    "Training Loss Curve"
)

plt.xlabel(
    "Iterations"
)

plt.ylabel(
    "Loss"
)

plt.grid()

plt.show()
```

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OutPut :-

```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
======== RESTART: C:/Users/Vinod Mali/Desktop/Ai/T092 Practical no 4.py ========
Dataset Loaded Successfully!

Dataset Shape:
(6607, 20)

First 5 Rows:
   Hours_Studied  Attendance  ...  Gender Exam_Score
0             23          84  ...    Male         67
1             19          64  ...  Female         61
2             24          98  ...    Male         74
3             29          89  ...    Male         71
4             19          92  ...  Female         70

[5 rows x 20 columns]

Missing values removed!

Classes:
0 = High
1 = Low
2 = Medium

Training Samples:
5102

Testing Samples:
```

```
Testing Samples:
1276

Neural Network trained successfully!

Accuracy:
0.9835423197492164

Accuracy Percentage:
98.35 %

Classification Report:
              precision    recall  f1-score   support

        High       0.00      0.00      0.00        10
         Low       0.64      0.54      0.58        13
      Medium       0.99      1.00      0.99      1253

    accuracy                           0.98      1276
   macro avg       0.54      0.51      0.52      1276
weighted avg       0.98      0.98      0.98      1276


Confusion Matrix:
[[   0    0   10]
 [   0    7    6]
 [   1    4 1248]]
```

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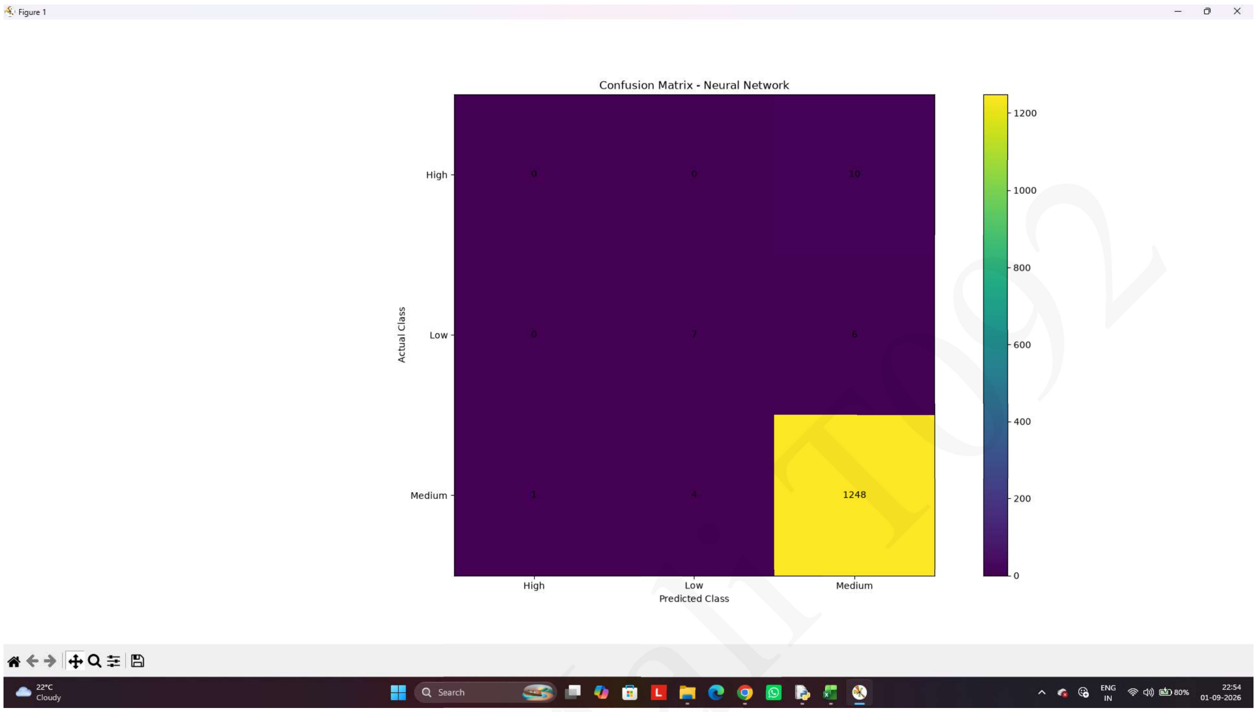
Graph :-



Observation :-

- **Confusion Matrix: Most predictions are correctly classified, showing good model performance.**
- **Training Loss Curve: The loss generally decreases as the number of iterations increases, indicating that the neural network is learning effectively.**
- **Overall Observation: The Feed Forward Backpropagation Neural Network provides satisfactory classification of student performance.**

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